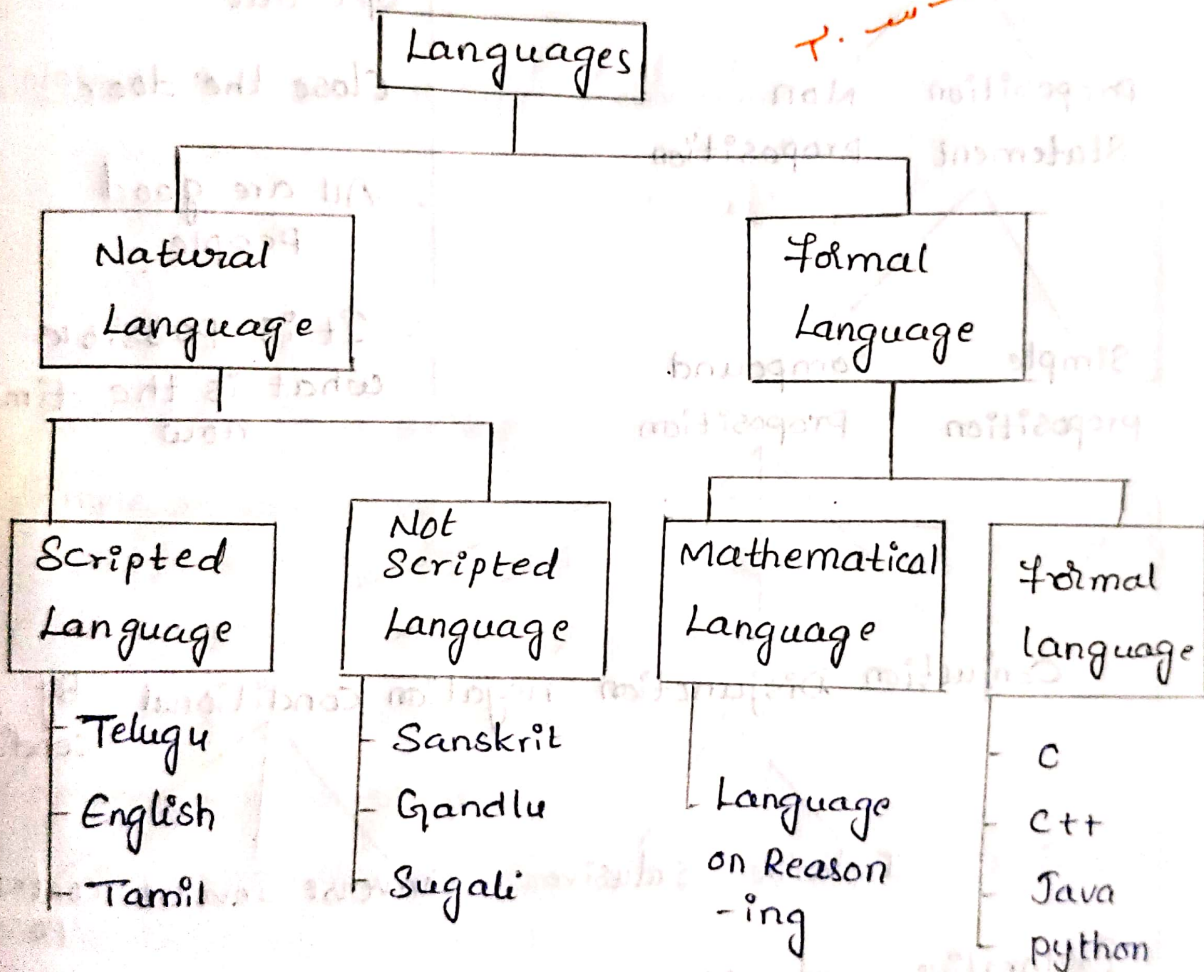


Logical Structures

①

Introduction - proposition and truth values
 tables - logical Connectives - Tautology and
 Contradiction - Logical Implication and
 equivalences - The algebra of propositions
 Arguments - formal proof - predicate Logic
 Theory of Inference.

Sentence:

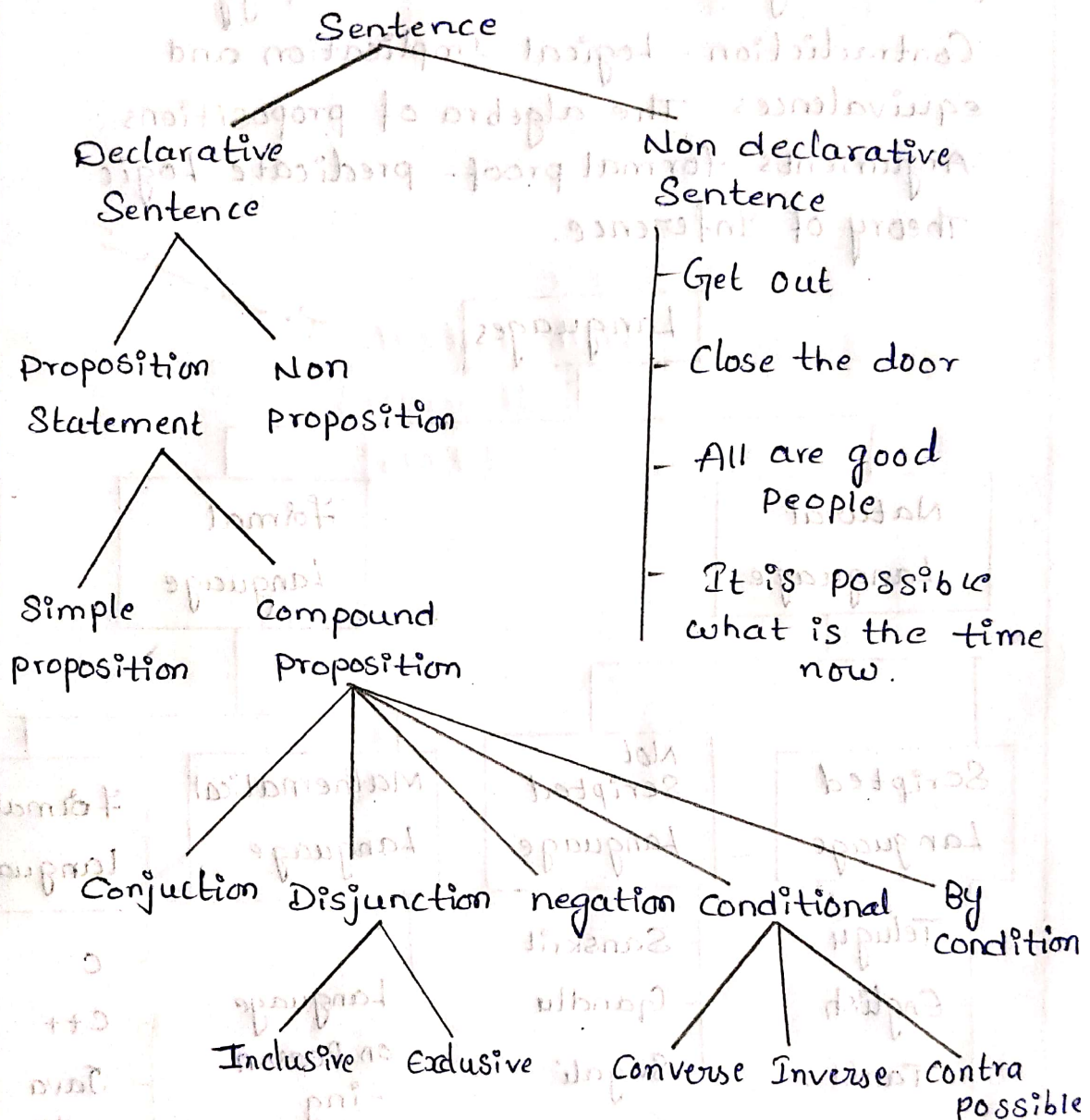
In any Scripted language the Sentence is the finite collection of words with specific Grammar rule.

- Ex: 1. Delhi is the Capital of India
 2. All roses are red
 3. Close the door.
 4. Get out.

5. what is the time now!

6. Is it possible

The following tree Structure illustrate the Classification of Sentence.



Declarative Sentence

A Sentence is called declarative if it is either true or false but not both.

Ex: Delhi is the Capital of India (True).

Proposition (Statement)

Any Sentence is proposition if it is declarative.

propositional (Statement)	Non-propositional (not a statement)
1. All roses are red (T)	1. All roses are beautiful
2. If $x > 1$ then $x = 100$ (T)	2. $x = 100$
3. $F = \frac{1}{2} mv^2$ (True)	3. $F = mv^2 + q$ (9)
4. $10 - 10 = 500$ (False)	4. Get out $\Rightarrow$ close the door $\Rightarrow$ what is the time now $\Rightarrow$ It is possible $\Rightarrow$ what is the nature?

Note: Any proposition is denoted

(i) $P, Q, R, S, \dots$

(ii) $P, Q, R, S, \dots$

(iii) $x, y, z, w, \dots$

(iv) $A, B, C, D, \dots$

Ex: $P: 2$ is a prime (True)

Simple proposition

A proposition is called simple proposition if it contain single statement or sentence.

Ex: $P: x + 2 = 500$ for $x > 1$

Compound proposition:

A proposition is called compound if it is contain more than one proposition.

Ex: $P: 1 - 1 = 10$ and all roses are red.

Truth Value and Truth Value Table

Let T denoted by True and F denoted by False then truth value of the given proposition 'F' is either T or F .

Ex: 1. The truth value of

$P: x^2 + y^2 = c^2 \forall x, y, c \in \mathbb{N}$ is "F"

Case (1):-

p is T and q is F then the following table shows that truth value table for p and q.

P	q
T	T
T	F
F	T
F	F

Case - (2):-

For three proposition P, q and r the truth value based table is

P	q	r
T	T	T
T	T	F
T	F	T
F	T	F
T	F	F
F	T	F
F	F	T
F	F	F

Logical Connectives (Operator)

- Negation
- Conjunction
- Disjunction
- Implication
- Bi-Implication

Negation

The negation of the statement P is denoted by $\sim P$ or $\bar{P}$ and defined by

$\sim P$: not P

Ex: $P : 1 - 5 = 500$

$\sim P : 1 - 5 \neq 500$

$P : \text{all roses are red}$

(5)

properties:

The truth value table for negation

P	$\sim P$
T	F
F	T

$\therefore \sim(\sim P) = P$

Conjunction:

The Conjunction of two given statements P and q is denoted $P \wedge q$ and defined

$P \wedge q : P \text{ and } q$

Ex: $P : 1 - 5 = 100$

$q : \text{All roses are red}$

$P \wedge q : 1 - 5 = 100 \text{ and All roses are red}$

properties:

$P \wedge \text{True} = P$

$P \wedge q = q \wedge P$

$(P \wedge q) \wedge r = P \wedge (q \wedge r)$

$\Rightarrow$ The truth table for $P \wedge q$ is

P	q	$P \wedge q$
T	T	T ✓
T	F	F
F	T	F
F	F	F

Disjunction

The disjunction of two given Statement P or q is denoted by $P \vee q$ and defined

$$P \wedge q : P \text{ or } q$$

Ex: $P = 1-5 = 100$

$q = \text{All roses are red}$

$$\therefore P \wedge q : 1-5 = 100 \text{ or all roses are red.}$$

Properties

$$P \vee P = P$$

$$P \vee q = q \vee P$$

$$(P \vee q) \vee r = P \vee (q \vee r)$$

P	q	$P \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Truth table for $P \vee q$

Implication [Conditional]

The Conditional of given Statement (proposition) of P and q is denoted by $P \rightarrow q$ and defined by

$$P \rightarrow q : \text{If } P \text{ then } q$$

Ex: P : It is raining

q : Delhi is the Capital of India.

$$P \rightarrow q : \text{If it is raining then delhi is the Capital of India.}$$

Properties

$$\Rightarrow P \rightarrow P \text{ is true}$$

$$\Rightarrow P \rightarrow q \neq q \rightarrow P$$

$$\Rightarrow P \rightarrow q \rightarrow r \neq P \rightarrow (q \rightarrow r)$$

Truth table for Conditional is

P	q	$P \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Bi-Conditional (or) Bi-Implication:

The Bi-Implication of given statements (proposition) of p and q is denoted by $p \leftrightarrow q$ is denoted or defined by 7

$$p \leftrightarrow q : p \text{ if and only if } q$$

Ex:

p : It is raining

q : Delhi is the capital of India

$p \leftrightarrow q$: It is raining if and only if Delhi is the capital of India's.

properties:

$\Rightarrow p \leftrightarrow q$ is true

$\Rightarrow p \leftrightarrow q = q \leftrightarrow p$

$\Rightarrow p \leftrightarrow q \leftrightarrow r = p \leftrightarrow (q \leftrightarrow r)$

$\Rightarrow$ The truth table for Bi-Conditional is

p	q	$p \leftrightarrow q$
T	T	T
T	F	F ✓
F	T	F ✓
F	F	T

Ex: Construct truth values tables for logical

formulas (i) $p \rightarrow (q \rightarrow r)$

(ii) $p \rightarrow (q \wedge r)$

(iii) $p \rightarrow (q \vee \bar{r})$

(i) The given logical formula is $p \rightarrow (q \rightarrow r)$

Truth table for given logical formula is.

P	q	r	$q \rightarrow r$	$P \rightarrow (q \rightarrow r)$
T	T	T	T	T
T	T	F	F ✓	F ✓
T	F	T	T	T
F	T	T	T	T
T	F	F	T	T
F	T	F	F ✓	T
F	F	T	T	T
F	F	F	T	T

ii) The given logical formula is $P \rightarrow (q \wedge r)$

P	q	r	$q \wedge r$	$P \rightarrow (q \wedge r)$
T	T	T	T	T
T	T	F	F	F
T	F	T	F	F
F	T	T	T	T
T	F	F	F	F
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T

iii) The given logical formula is

$$P \rightarrow (q \vee \bar{r})$$

Truth table for given logical formula is

P	q	r	$\bar{r}$	$q \vee \bar{r}$	$P \rightarrow (q \vee \bar{r})$
T	T	T	F	T	T
T	T	F	T	T	T
T	F	T	F	F	F
F	T	T	F	T	T
T	F	F	T	T	T
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	T	T

Construct truth tables for: (i) $P \rightarrow (q \wedge r)$
(ii) $(\bar{p} \vee q) \leftrightarrow \bar{r}$.

(i) The given logical formula is $P \rightarrow (q \wedge r)$

P	q	r	$q \wedge r$	$P \rightarrow (q \wedge r)$
T	T	T ✓	T	T
T	T	F	F	F
T	F	T	F	F
F	T	T	T	T
T	F	F	F	F
F	T	F	F	T
F	F	T	F	T
F	F	F ✓	F	T

(ii) The given logical formula is $(\bar{p} \vee q) \leftrightarrow \bar{r}$

P	q	r	$\bar{r}$	$\bar{p}$	$\bar{p} \vee q$	$(\bar{p} \vee q) \leftrightarrow \bar{r}$
T	T	T	F	F	T	F ✓
T	T	F	T	F	T	T
T	F	T	F	F	F	T
F	T	T	F	T	T	F ✓
T	F	F	T	F	F	F ✓
F	T	F	T	T	T	T
F	F	T	F	T	T	F ✓
F	F	F	T	T	T	T

2. Use truth value tables to establish
- (i) $A \rightarrow B$ and $A \leftrightarrow B$ when (ii) $x \wedge y$ and $x \vee y$
- $x: P \rightarrow (Q \rightarrow P)$
- $A: P \wedge Q \rightarrow (Q \wedge \bar{P})$ and $y: Q \rightarrow (P \rightarrow Q)$
- $B: P \wedge Q \rightarrow (\bar{Q} \wedge P)$. 10

(i) The given logical formula is $(P \wedge Q) \rightarrow (Q \wedge \bar{P})$

P	Q	$P \wedge Q$	$\bar{P}$	$Q \wedge \bar{P}$	$(P \wedge Q) \rightarrow (Q \wedge \bar{P})$
T	T	T	F	F	F ✓
T	F	F	F	F	T
F	T	F	T	T ✓	T
F	F	F	T	F	T

The given logical formula is $P \wedge Q \rightarrow (\bar{Q} \wedge P)$

P	Q	$P \wedge Q$	$\bar{Q}$	$\bar{Q} \wedge P$	$P \wedge Q \rightarrow (\bar{Q} \wedge P)$
T	T	T	F	F	F ✓
T	F	F	T	T ✓	T
F	T	F	F	F	T
F	F	F	T	F	T

Table for $A \rightarrow B$ and $A \leftrightarrow B$

A	B	$A \rightarrow B$	$A \leftrightarrow B$
F	F	T	T
T	T	T	T
T	F	F	F
F	T	F	F

3. Given the three propositions P, Q, and R
- Construct truth tables for:

- (i) $(P \wedge Q) \rightarrow \bar{r}$ (iii) $P \wedge (\bar{Q} \wedge r)$
- (ii) $(P \vee r) \wedge \bar{Q}$ (iv) $P \rightarrow (\bar{Q} \vee \bar{r})$
- (v) $(P \vee Q) \leftrightarrow (r \vee P)$

(i) Given logical formula is $(P \wedge q) \rightarrow \bar{r}$

P	q	r	$P \wedge q$	$\bar{r}$	$(P \wedge q) \rightarrow \bar{r}$
T	T	T	T	F	T ✓
T	T	F	T	T	F
T	F	T	F	F	F
F	T	T	F	F	F
T	F	F	F	T	F
F	T	F	F	T	F
F	F	T	F	F	F
F	F	F	F	T	F

Tautology and Contradictions

A logical formula is called a tautology if it's last column of truth tables contains all components are "true (T)".

Otherwise logical formula is called Contradiction.

A logical formula is called a Contingency if it's last column of the truth tables contains "True and False".

If Contingencies is neither tautology nor Contradiction

Ex:-

P	$\sim P$	$P \vee \sim P$	$P \wedge \sim P$	$P \rightarrow \sim P$	$P \leftrightarrow \sim P$
T	F	T	F	F	F
F	T	T	F	T	F

$\Rightarrow P \vee \sim P$ is a tautology

$\Rightarrow P \wedge \sim P$ and $P \leftrightarrow \sim P$ are both Contradiction

$\Rightarrow P \rightarrow \sim P$ is a Contingency.

4 (i) Determine whether each of the following is a tautology, a Contradiction or neither.

(i) $P \rightarrow (P \vee q)$

P	q	$P \vee q$	$P \rightarrow (P \vee q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

(12)

Here

From the above table last column contains all are "true". Hence the given logical formula is a tautology.

Logically Equivalent

Two logical formulas L and L' are called logically equivalent if it is denoted by $L \equiv L'$ ($L \leftrightarrow L'$) and defined as all components in the last column of the truth table of L is the same as L' .

Example:-

$P \rightarrow q$ and $\sim P \vee q$ are equivalent.

P	q	$P \rightarrow q$
T	T	T
T	F	F ✓
F	T	T
F	F	T

P	q	$\sim P$	$\sim P \vee q$
T	T	F	T
T	F	F	F ✓
F	T	T	T
F	F	T	T

Hence,

$$P \rightarrow q \equiv \sim P \vee q$$

Show that the biconditional proposition $P \leftrightarrow q$ is logically equivalent to the conjunction of the two conditional proposition $P \rightarrow q$ and $q \rightarrow P$.

$$P \leftrightarrow q \equiv (P \rightarrow q) \wedge (q \rightarrow P)$$

(or)

$$P \leftrightarrow q \equiv (\sim P \vee q) \wedge (\sim q \vee P)$$

The truth table for $P \leftrightarrow Q$

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F ✓
F	T	F ✓
F	F	T

The truth table for $(P \rightarrow Q) \wedge (Q \rightarrow P)$

P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \wedge (Q \rightarrow P)$
T	T	T	T	T
T	F	F	T	F ✓
F	T	T	F	F ✓
F	F	T	T	T

From the above two tables, we conclude that

$$P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$$

6. Verify that $(P \leftrightarrow Q)$

(i) $(P \wedge Q)$ and $(P \rightarrow \bar{Q})$ (ii) $(P \leftrightarrow Q) \wedge Q$ and P are logically equivalent or not.

(i) The truth table for $(P \wedge Q) \wedge P \rightarrow \bar{Q}$

P	Q	$P \wedge Q$	P	Q	$\bar{Q}$	$P \rightarrow \bar{Q}$
T	T	T ✓	T	T	F	F ✓
T	F	F	T	F	T	T
F	T	F	F	T	F	T
F	F	F	F	F	T	T

(ii) The truth table for $(P \leftrightarrow Q) \wedge Q$

P	Q	$P \leftrightarrow Q$	$(P \leftrightarrow Q) \wedge Q$	P
T	T	T	T ✓	T
T	F	F ✓	F	T
F	T	F ✓	F	F
F	F	T	F	F

Here the above two tables are all the components of the last column are not equal.

Hence, the given two logical equivalents are not equivalent. 14

$$P \wedge q \not\equiv P \leftrightarrow \bar{q}$$

From the above table 1st and last column components are not equivalent

And Hence.

$$(P \leftrightarrow q) \not\equiv P$$

Algebraic properties of Logical formulas.

(Algebraic Method) ✓

1. Identity Law:

$$P \wedge P \equiv P \text{ and } P \vee P \equiv P$$

2. Commutative Law

$$P \wedge q \equiv q \wedge P \text{ and } P \vee q \equiv q \vee P$$

3. Associative Law

$$(P \wedge q) \wedge r \equiv P \wedge (q \wedge r) \text{ and } (P \vee q) \vee r \equiv P \vee (q \vee r)$$

4. Distributive Law

$$P \wedge (q \vee r) \equiv (P \wedge q) \vee (P \wedge r)$$

$$P \vee (q \wedge r) \equiv (P \vee q) \wedge (P \vee r)$$

5. De Morgan Law

$$\sim(P \wedge q) \equiv \sim P \vee \sim q$$

$$\sim(P \vee q) \equiv \sim P \wedge \sim q$$

6. Double Negation Law

$$\sim(\sim P) \equiv P$$

7. Inverse Law

$$P \vee \sim P \equiv T$$

$$P \wedge \sim P \equiv F$$

8. Absorption Law

$$P \wedge (P \vee Q) \equiv P$$

$$P \vee (P \wedge Q) \equiv P$$

Domination Law

$$P \wedge F \equiv F$$

$$P \vee T \equiv T$$

Complement Law

$$P \wedge \sim P \equiv F$$

$$P \vee \sim P \equiv T$$

9. Explain the algebra of propositions. Prove each of the following logical equivalences using the method of algebra of propositions.

(i) $(\bar{P} \wedge Q) \vee (\bar{P} \vee Q) \equiv \bar{P}$ (ii) $(P \wedge \bar{P}) \vee (P \vee \bar{P}) \equiv T$

(iii) $(P \wedge Q) \wedge Q \equiv P \wedge Q$ (iv) $P \rightarrow Q \equiv \bar{P} \vee Q$

(v) $(P \wedge Q) \rightarrow R \equiv (\bar{P} \vee \bar{Q}) \vee R$

(vi) The given logical formula is $(\bar{P} \wedge Q) \vee (\bar{P} \vee Q) \equiv \bar{P}$

Truth table method

P	Q	$\bar{P}$	$\bar{Q}$	$\bar{P} \wedge Q$	$P \vee Q$	$\bar{P} \vee Q$	$\bar{P} \wedge Q \vee (\bar{P} \vee Q)$
T	T	F	F	F	T	F	F
T	F	F	T	F	T	F	F
F	T	T	F	T	T	F	T
F	F	T	T	F	F	T	T

From the above table 3rd and last column are same. Hence $(\bar{P} \wedge Q) \vee (\bar{P} \vee Q) \equiv \bar{P}$.

Algebraic Method

$$\text{LHS} = (\bar{P} \wedge Q) \vee (\bar{P} \vee Q)$$

$$= (\bar{P} \wedge Q) \vee (\bar{P} \wedge \bar{Q}) \text{ by De Morgan's law.}$$

$$= \bar{P} \wedge (Q \vee \bar{Q}) \text{ by distributive law}$$

$$= \bar{P} \wedge T \text{ by Complement law}$$

$$\text{RHS} = \bar{P} \text{ by identity law.}$$

Absorption Laws

$$P \wedge (P \vee Q) \equiv P$$

$$P \vee (P \wedge Q) \equiv P$$

Idempotent Law

$$P \wedge P \equiv P$$

$$P \vee P \equiv P$$

(ii) $(P \wedge \bar{P}) \vee (\bar{P} \vee P) \equiv T$

Truth table

P	$P \wedge \bar{P}$	$\bar{P}$	$\bar{P} \vee P$	$(P \wedge \bar{P}) \vee (\bar{P} \vee P)$
T	F	F	F	F
F	F	T	T	T

From the above table, we have

$$(P \wedge \bar{P}) \vee (\bar{P} \vee P) \equiv T$$

Algebraic Method

$$\text{RHS} = (P \wedge \bar{P}) \vee (\bar{P} \vee P)$$

$$= P \vee \bar{P} \text{ by Idempotent law, } P \wedge \bar{P} \equiv F$$

$$= T \text{ by Inverse law, } \bar{P} \vee P \equiv T$$

$$= \text{RHS}$$

(iii) $(P \wedge Q) \wedge Q \equiv P \wedge Q$

Truth table

P	$P \wedge Q$	Q	$(P \wedge Q) \wedge Q$
T	T	T	T
T	F	F	F
F	F	T	F
F	F	F	F

From the above table, we have

$$(P \wedge Q) \wedge Q \equiv P \wedge Q$$

Algebraic Method

$$\text{RHS} = (P \wedge Q) \wedge Q$$

$$\equiv P \wedge (Q \wedge \bar{Q}) \text{ by associative law.}$$

$$= P \wedge Q, \text{ by Idempotent law.}$$

$$= \text{RHS.}$$

$$P \rightarrow Q \equiv \overline{P \wedge \bar{Q}}$$

Truth table.

P	Q	$\bar{Q}$	$P \wedge \bar{Q}$	$P \rightarrow Q$	$\overline{P \wedge \bar{Q}}$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	F	T	T
F	F	T	F	T	T

From the above table
5th column and last column are equal.

$$P \rightarrow Q \equiv \overline{P \wedge \bar{Q}}$$

Algebraic method

$$\text{LHS} = P \rightarrow Q$$

$$\text{RHS} = \overline{P \wedge \bar{Q}}$$

$$\equiv \bar{P} \vee (\bar{\bar{Q}}) \text{ by demorgan's law}$$

$$\equiv \bar{P} \vee Q, \text{ by double negation.}$$

$$\equiv P \rightarrow Q, \text{ since } (P \rightarrow Q \equiv \bar{P} \vee Q)$$

$$= \text{LHS.}$$

$$(P \wedge Q) \rightarrow R \equiv (\bar{P} \vee \bar{Q}) \vee R$$

Truth table.

P	Q	R	$\bar{P}$	$\bar{Q}$	$P \wedge Q$	$\bar{P} \vee \bar{Q}$	$(\bar{P} \vee \bar{Q}) \vee R$	$P \wedge Q \rightarrow R$
T	T	T	F	F	T	F	T	T
T	F	F	F	T	F	T	T	F
T	F	T	F	T	F	T	T	T
F	T	T	T	F	F	T	T	T
T	T	F	F	F	T	F	F	F
F	T	F	T	F	F	T	T	T

F	F	T	T	T	F	T	T	T
F	F	F	T	T	F	T	T	T

From the above table

$$(p \wedge q) \rightarrow r \equiv (\bar{p} \vee \bar{q}) \vee r$$

(12)

Algebraic method

$$\text{LHS} = (p \wedge q) \rightarrow r$$

$$\equiv \sim(p \wedge r) \vee r, \text{ Since}$$

$$A \rightarrow B \equiv \sim(A \vee B)$$

$$\equiv (\sim p \vee \sim q) \vee r$$

$$\equiv (\bar{p} \vee \bar{q}) \vee r$$

$$= \text{RHS.}$$

$$\text{RHS} = (\bar{p} \vee \bar{q}) \vee r$$

$$\equiv (\overline{p \wedge q}) \vee r$$

$$\equiv \sim(p \wedge q) \rightarrow r$$

$$= (p \wedge q) \rightarrow r$$

$$= \text{LHS.}$$

Use the method The algebra of propositions to show that $p \wedge (q \vee \bar{p})$ is logically equivalent to $p \wedge q$.

The given logical formula $p \wedge (q \vee \bar{p}) \equiv p \wedge q$

Truth table

p	q	$\bar{p}$	$q \vee \bar{p}$	$p \wedge (q \vee \bar{p})$	$p \wedge q$
T	T	F	T	T ✓	T ✓
T	F	F	F	F	F
F	T	T	T	F	F
F	F	T	T	F	F

From the above table we have,

$$\boxed{p \wedge (q \vee \bar{p}) \equiv p \wedge q}$$

Algebraic Method

$$\text{LHS} = P \wedge (q \vee \bar{p})$$

$$= (P \wedge q) \vee (P \wedge \bar{p}) \text{ by Distributive law}$$

$$= (P \wedge q) \vee \underline{F} \text{ by Complementary law, } p \vee \bar{p} \equiv F$$

$$= P \wedge q \text{ by Identity Law}$$

$$= \text{RHS.}$$

$$A \vee F \equiv A$$

Premises:

In mathematical logic any assumption any axiom, any hypothesis. any logical statement is called premises.

Example:

* Suppose x is a number $\rightarrow$ Assumption

* $a+b = b+a$ if $a, b \in \mathbb{N}$ (Axiom)

* Given that p is a prime (hypothesis).

* $P \wedge q$

$P \vee q$

$P \rightarrow q$

$P \leftrightarrow q$

$P \wedge q \rightarrow r$

logical statements.

Inference theory

Let $P_1, P_2, \dots, P_n$ be the finite collection of premises and let 'C' be the Conclusion. A mathematical procedure for constructing 'C' for given premises is called inference theory.

Method for Inference

- $\rightarrow$ Truth table method for inference
- $\rightarrow$ Derivations of logic method for inference.

Truth table method for Inference

Step:1

$P_1, P_2, \dots, P_n$ be the given premises.

and C be the given Conclusion.

(20)

Step:2

Construct Conjunction $P_1, P_2, \dots, P_n$

$$P_1 \wedge P_2 \wedge \dots \wedge P_n$$

Step:3

Develop the Conditional with $P_1 \wedge P_2 \wedge \dots \wedge P_n$ and Conclusion ' C ' as

$$(P_1 \wedge P_2 \wedge \dots \wedge P_n) \rightarrow C$$

Step:4

If $\odot$ is a tautology then C is the Conclusion of required premises otherwise it is ~~is~~ invalid.

Example:

Show that the premises Q leads to the Conclusion $P \vee Q$ by using truth table.

Premises : Q

Conclusion : $P \vee Q$

Conditional

$$Q \rightarrow P \vee Q$$

P	Q	$P \vee Q$	$Q \rightarrow P \vee Q$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	T

From the above table, we observe that

$Q \rightarrow P \vee Q$, is a tautology, and hence the premises

Q leads to the conclusion

Test the validity of the following argument.

If you insulted Bob, then I'll never speak to you again. You insulted Bob, so I'll never speak to you again. (21)

The given statemental formula is

"If you insulted bob, then
I'll never speak to you"

again, you insulted bob, so

I will never speak to you again.

Taking P : You insulted bob

q : I will ^{never} speak to you.

premises:

$P \rightarrow q$ and P

Conclusion:

q

Conditional

$(P \rightarrow q) \wedge P \rightarrow q$

P	q	$P \rightarrow q$	$(P \rightarrow q) \wedge P$	$(P \rightarrow q) \wedge P \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

From the above table, we observe that $(P \rightarrow q) \wedge P \rightarrow q$ is a tautology, and hence the given statement is valid.

12. Test the validity of the following argument.
 If you are a mathematician, then you are clever.
 You are clever and rich. Therefore, if you are rich,
 then you are mathematician. (21)

Sol:-

The given Statemental formula is

If you are a mathematician, then you are clever.
 You are clever and rich. Therefore, if you are rich
 then you are mathematician. ^{Hence, so} Conclusion

Taking : P : "you are a mathematician"

Q : "You are clever"

R : "You are rich"

Premises: $P \rightarrow Q$ and $Q \wedge R$.

Conclusion: $R \rightarrow P$

Condition: $(P \rightarrow Q) \wedge (Q \wedge R) \rightarrow (R \rightarrow P)$

$(P \rightarrow Q) \wedge (Q \vee R) \rightarrow (R \rightarrow P)$

Truth table

P	Q	R	$P \rightarrow Q$	$Q \wedge R$	$(P \rightarrow Q) \wedge (Q \wedge R)$	$R \rightarrow P$	$(P \rightarrow Q) \wedge (Q \wedge R) \rightarrow (R \rightarrow P)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	T	T
T	F	T	F	F	F	T	T
F	T	T	F	T	F	T	F ✓
F	F	T	T	F	F	F	F
F	T	F	T	F	F	T	T
F	F	T	T	F	F	F	T
F	F	F	T	F	F	T	T

From the above table we observe that,
 the Conditional $(P \rightarrow Q) \wedge (Q \vee R) \rightarrow (R \rightarrow P)$ is not
 a tautology. By the theory of inference the
 given statemental formula is invalid.

Rules for Inference (Inference using Derivation)
 Without constructing the truth table, it is possible to determine whether the given premises through a valid conclusion. This mathematical conclusion procedure is called Inference using Derivation. The process of derivation, the following rules are applied systematically.

Rule P:

A premises may be introduced at any point in the derivation.

Rule T:

A tautologically implied formula may be introduced at any point in the derivation.

In this rule T, the implied formulas are listed below.

Modus Ponens:

Premises

$$\begin{array}{l} P \rightarrow Q \\ P \end{array}$$

Conclusion:

$$Q$$

Modus Tollens:

Premises:

$$\begin{array}{l} P \rightarrow Q \\ \sim Q \end{array}$$

Conclusion

$$\sim P$$

Disjunctive syllogism

Premises

$$\begin{array}{l} P \vee Q \\ \sim P \end{array}$$

Conclusion

$$Q$$

Hypothetical syllogism

Premises

$$\begin{array}{l} P \rightarrow Q \\ Q \rightarrow R \end{array}$$

Conclusion

$$P \rightarrow R$$

In the theory of inference, following table describes most important Inference rules with their conclusion.

Name of the Rule	Premises $\implies$	Conclusion
Simplification (Simp)	$P \wedge Q$	P
Addition (Add)	P	$P \wedge Q$
Conjunction (Conj)	P, Q	$P \wedge Q$
Disjunctive Syllogism (DS)	$P \vee Q, \sim P$	Q
Modus ponens MP	$P, P \rightarrow Q$	Q
Modus tollens MT	$P \rightarrow Q, \sim Q$	$\sim P$
Hypothetical Syllogism HS	$P \rightarrow Q, Q \rightarrow R$	$P \rightarrow R$
Absorption Abs	$P \rightarrow Q$	$P \rightarrow (P \wedge Q)$
Constructive Dilemma CD	$P \rightarrow Q, R \rightarrow S$ $P \vee R$	$Q \vee S$

Rule CP: [Rule of Conditional proof]

If, under the assumption that the logical form A is true, you can derive B (i.e., assume B), then we conclude the Conditional $A \rightarrow B$ exists.

Example

Show that $R \wedge (P \vee Q)$ is a valid Conclusion from the premises. $P \vee Q, Q \rightarrow R, P \rightarrow M$ and $\sim M$.

sol

Premises:

$$P \vee Q$$

$$Q \rightarrow R$$

$$P \rightarrow M$$

$$\sim M$$

Conclusion

$$R \wedge (P \vee Q)$$

Step: 1 $P \rightarrow M$ (By Rule P)

Step: 2 $\sim M$ (By Rule P)

Step:3 : $\sim P$ By Step (1) & (2) and modus Tollens
[By Rule P]

Step:4 : $P \vee Q$ [By rule P]

Step:5 : Q [By Rule T and Disjunctive Syllogism
using Step 3 & 4]

Step:6 : $Q \rightarrow R$ [By rule P]

Step:7 : R [By Rule T and modus ponens using
Step 5 & 6]

Step:8 : $R \wedge (P \vee Q)$ [By Rule T and Conjunction
using Step 6 & 7]

13) Explain the rules of premises.

i) Construct a formal proof of the validity of the
following argument. premises: $P \rightarrow q, q \wedge r$
Conclusion: q .

Premises

Conclusion

$P \rightarrow q, q \wedge r$

Step 1 : $P \wedge R$ [By rule P]

Step 2 : P [By Rule T and Simplification using Step (1)]

Step 3 : $P \rightarrow Q$ [By Rule P]

Step 4 : Q [By Rule T and modus ponense using
Step 2 & 3]

* Show that $S \vee R$ is a tautologically implied by
 $(P \vee Q) \wedge (P \rightarrow Q) \wedge (Q \rightarrow S)$ by using inference
theory.

The given logical formula is

$(P \vee Q) \wedge (P \rightarrow Q) \wedge (Q \rightarrow S)$.

Consider the following premises

$P \vee Q$

$P \rightarrow Q$

$Q \rightarrow S$

Conclusion : $S \vee R$

13
(iii)

Construct a formal proof of the following argument.
premises: $P \rightarrow Q, R \rightarrow S, \bar{Q}, R$ Conclusion: $P \rightarrow R \wedge S$

Sol

premises:

$P \rightarrow Q, R \rightarrow S, \bar{Q}, R$

Conclusion:

$P \wedge S$

(2)

Step 1: $P \rightarrow Q$ (By Rule P)

Step 2: $\bar{Q}$ (By Rule P)

Step 3: $\bar{P}$ [By Rule T and Modus Tollens using Step (1) & (2)]

Step 4: $R \rightarrow S$ [By Rule P]

Step 5: R [By rule P]

Step 6: S By rule T and modus ponens using Step 4 and (5)

Step 7: $P \wedge S$ (By Rule T and Conjunction using Step 5 & 6)

(iv)

Give a formal proof for the following argument:

premises: $P \wedge (Q \vee R), \bar{P} \vee \bar{Q}$ Conclusion: $R \vee Q$

Sol

Given premises

Conclusion

$P \wedge (Q \vee R),$

$R \vee Q$

$\bar{P} \vee \bar{Q}$

Step 1: $P \wedge (Q \vee R)$ By Rule P

Step 2: P [By Rule T and Simplification using Step 1]

Step 3: $\bar{P} \vee \bar{Q}$ (By rule P)

Step 4: $\bar{Q}$ By Rule and disjunctive syllogism using Step (2) & (3)

Step 5: $(Q \vee R) \wedge P$ (By Rule T and Commutativity ($A \wedge B \equiv B \wedge A$) using Step 1)

$\bar{P}, \bar{P} \vee \bar{Q} \Rightarrow \bar{Q}$
Replace
 P by $\bar{P}$ We get
 $P, \bar{P} \vee \bar{Q} \Rightarrow \bar{Q}$

Step 6: $q \vee r$ By Rule T and Simplification using Step 4

Step 7: r By rule T and disjunctive Syllogism using Step (4) & (5). (27)

Step 8: $r \vee q$ By Rule T and addition using Step (q) with $(\bar{q}) = q$.

ii) provide a formal proof of the validity of the following argument. Premises: $P \rightarrow q, \bar{q} \vee r$
Conclusion: $P \rightarrow r$. [Based on Rule CP]

Given premises.

$$P \rightarrow q, \bar{q} \vee r$$

Conclusion:

$$P \rightarrow r$$

Step 1: $P \rightarrow q$ [By Rule P]

Step 2: P [By rule CP and assumed premises]

Step 3: q By Rule T and modus ponens using Step (1) & (2).

Step 4: $\bar{q} \vee r$ [By rule P]

Step 5: r By rule T and Disjunctive syllogism using Step (3) & (4)

$\bar{P}, P \vee q \Rightarrow q$
Replace P by $\bar{P}$
$P, \bar{P} \vee q \Rightarrow q$

Step 6: $P \rightarrow r$ By rule CP using Step (2) & (5).

Predicates:

A predicate is a mathematical statement which contains an unspecified mathematical symbol. This unspecified symbol is called variable.

Suppose the variable is denoted by x and x is in D . Then D is called domain of discourse.

Ex: $x^2 < x + 1$
 $x^2 + 2x - 1 > 0$

* x is a multiple of 2.

These are all predicates of the variable x .
For any variable x the predicate represented by $P(x)$. Similarly for any two variable x and y the predicate is represented by $P(x, y)$. Generally for n variables the predicate represented by $P(x_1, x_2, x_3, \dots, x_n)$ (28)

Ex:-

- * $P(x) : x^2 < x + 1$
- * $P(x, y) : x^2 + y^2 = 5^2$
- * $P(x, x_2, \dots, x_n) : 2x_1 - 5x_2^2 + x_3^2 - \dots + 2x_n = 1$

Properties of predicates

1. Every predicate is not a proposition but it is a mathematical expression.
2. The variable x in the predicate $P(x)$ is always a free variable.
3. Every predicate $P(x)$ indicate the Characteristic of given object x .
4. predicates are fundamental tools in mathematical reasoning and intelligent system.
5. In any object oriented language, each and every statement contains predicates.

Quantifiers

Quantifiers are mathematical symbol which are used in predicate calculus to express and develop the scope of new variables with given data.

Consider the following list of propositions:

1. All people are mortal.
2. Every computer is a 16-bit machine.

3. No birds are black
4. Some people have blue eyes.
5. There exists an even prime number.

From the above Statement, each Statement Contains a special word. namely all, every, None, Some, and one. These special type of words are called Quantifiers in the predicate logic.

The following classification represents the types of Quantifiers.

Two types:

1. Universal Quantifier [All, Every]
2. Existential Quantifier [None, Some, one]

1. Universal Quantifier

The universal Quantifier of the given data is denoted by the mathematical Symbol " $\forall$ " and read as "every", "for all" for any variables. x the universal quantifier of the predicate. $P(x)$ is represented by

$$\forall x(P(x))$$

2. Existential Quantifier

The existential quantifier is denoted by the mathematical Symbol " $\exists$ " and read as for. Some, there exists. For any variable x the existential quantifier of the predicate $P(x)$ is represented by

$$\exists x(P(x))$$

14. Describe the terminology of predicates and quantifiers. Symbolise the propositions:
- (i) Every day I go jogging.
 - (ii) Some people think of no one but themselves.
 - (iii) Some of the children didn't apologise.
 - (iv) Nobody like cheats.

Sol
(i)

The given statemental formula is

"Every ^{domain} day I go jogging" predicate (30)

Domain of discourse

Days(x)

predicates

D(x) : x is a day

J(x) : I go jogging on day

Representation of Quantifier

$$\forall x (D(x) \rightarrow J(x))$$

(ii)

The given statemental formula is

"Some people think of no one but themselves".

Domain of discourse

People(x)

predicates

P(x) : x is a people

N(x) : x thinks of no one but themselves.

Representation of Quantifier

$$\exists x (P(x) \wedge N(x))$$

(iii)

So The given statemental formula is

"Some of the children didn't apologise"

Domain of Discourse

Children(x)

predicates

C(x) : x is a children

A(x) : x didn't is apologise

Representation of Quantifier

$$\exists x (C(x) \wedge \sim A(x))$$

(iv)

The given statemental formula is

"Nobody likes cheats".

Domain the discourse

people

Predicates

P(x) : x is a people

C(x) : x is cheats.

$$\sim \exists x (P(x) \wedge C(x))$$

Inference theory in predicate logic

The following table describes the different types of Inference for deriving many predicates on their corresponding Quantifier. (31)

Name of the rule	Premises	Conclusion
Universal specification	$\forall x(P(x))$	$P(y)$
Existential specification	$\exists x(P(x))$	$P(y)$
Universal Generalization	$P(y)$	$\forall x(P(x))$
Existential Generalization	$P(y)$	$\exists x(P(x))$
Modus ponens	$P(x), P(x) \rightarrow Q(x)$	$Q(x)$
Modus tollens	$P(x) \rightarrow Q(x), \sim Q(x)$	$\sim P(x)$
Universal Modus ponens	$P(x), \forall x(P(x) \rightarrow Q(x))$	$Q(y)$
Universal Modus tollens	$\forall x(P(x) \rightarrow Q(x)), \sim Q(x)$	$\sim P(y)$

Q Explain Arguments, specifications, and Generalizations in predicate logic.

Q Show that the following is a valid argument:
 "Everyone who has green eyes is not to be trusted.
 Bill has green eyes. Therefore, Bill is not to be trusted."

The given statemental formula is
 "Everyone who has green eyes is not to be trusted. Bill has green eyes. Therefore, Bill is not to be trusted."

Domain

People (x)

Predicates:

$G(x)$: x has green eyes

$T(x)$: x is to be trusted

b : Bill

Premises

$\forall x (G(x) \rightarrow \sim T(x))$

$G(b)$

(B1)

Conclusion:

$\sim T(b)$

Step 1: $\forall x (G(x) \rightarrow \sim T(x))$ (By Rule P)

Step 2: $G(b) \rightarrow \sim T(b)$ (By Rule T and Universal Specification using step(1))

Step 3: $G(b)$ (By Rule P)

Step 4: $\sim T(b)$ (By Rule T and modus Tollense using Step(2) & (3))

"All student go to parties. some students drink too much
Domain Therefore. Some people who Premises

Students (x) drink too much $\forall x (S(x) \rightarrow P(x))$
go to parties $\exists x (S(x) \wedge D(x))$

Predicates

$S(x)$: x is a Student

$P(x)$: x goes to parties

$D(x)$: x drinks too much

Conclusion:

$\exists x (P(x) \wedge D(x))$

Step(1): $\forall x (S(x) \wedge D(x))$ (By rule P)

Step(2): $S(a) \wedge D(a)$ (By rule T and Existence Spec

Step(3): $S(a)$ (By Rule T and Simplification)

Step(4): $\forall x (S(x) \rightarrow P(x))$ (By rule P)

Step(5): $S(a) \rightarrow P(a)$ (Rule T and Universal Specific

Step(6): $P(a)$ (By Rule T and modus ponense)

Step(7): $D(a) \wedge S(a)$ (By Rule T and Commutativity)

Step(8): $D(a)$ (By Rule T and Simplification)

Step(9): $D(a) \wedge P(a)$ (By Rule T and Conjunctive)

Step(10): $\exists x (D(x) \wedge P(x))$ (By Rule T Existential generalization)

See
Back
side

(iii) "Everyone shouts or cries. Not everyone cries. So Some people shout and don't cry." (33)

Domain

People(x)

Predicates

$S(x)$: x shouts

$C(x)$: x cries.

Premises

$\forall x (S(x) \vee C(x))$
 $\sim \forall x C(x)$

Conclusion

$\exists x (S(x) \wedge \sim C(x))$

Step(1): $\forall x (S(x) \vee C(x))$ (By Rule P)

Step(2): $S(a) \vee C(a)$ (By Rule T and Universal Specification)

Step(3): $\sim \forall x C(x)$ (By Rule P)

Step(4): $\exists x (\sim C(x))$ (By Rule T and property)

Step(5): $\sim C(a)$ (By Rule T and Existential Specification)

Step(6): $S(a)$ (By Rule T and Disjunctive syllogism)

Step(7): $S(a) \wedge \sim C(a)$ (By Rule T and conjunctive)

Step(8): $\exists x [S(x) \wedge \sim C(x)]$ (By Rule T and Existential Generalization)